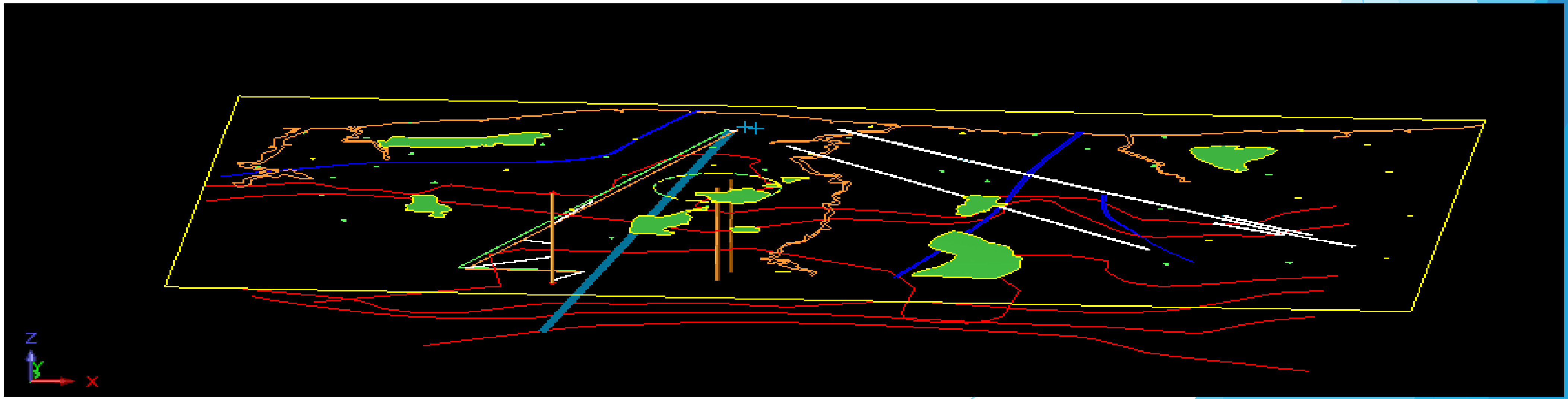
# Mine Planning-Introduction

B S Choudhary

Department of Mining Engineering

IIT(ISM) Dhanbad



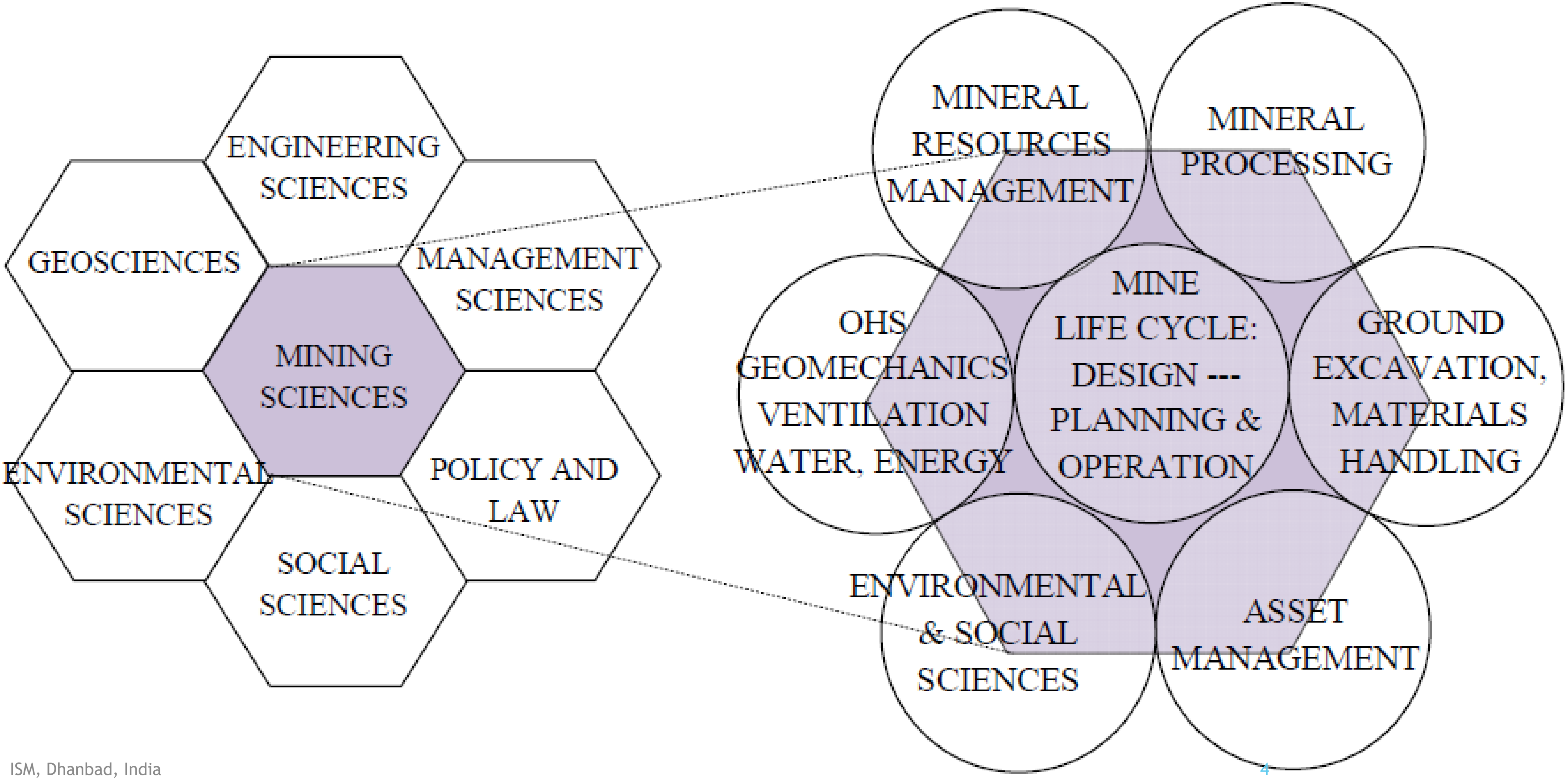
# Introduction

- Mine planning and its importance;
- Components of mine planning,
- Technical and economic information for planning,
- Exploratory drills and
- Interpretation of bore hole data

# Mining that is

- Excavation made on the surface or underneath the surface for excavation of mineral
- financially viable;
- socially responsible;
- environmentally, technically and scientifically sound; with a long term view of development;
- uses mineral resources optimally; and,
- ensures sustainable post-closure land uses.
- Also one based on creating long-term, genuine, mutually beneficial partnerships between government, communities and miners, based on integrity, cooperation and transparency”.

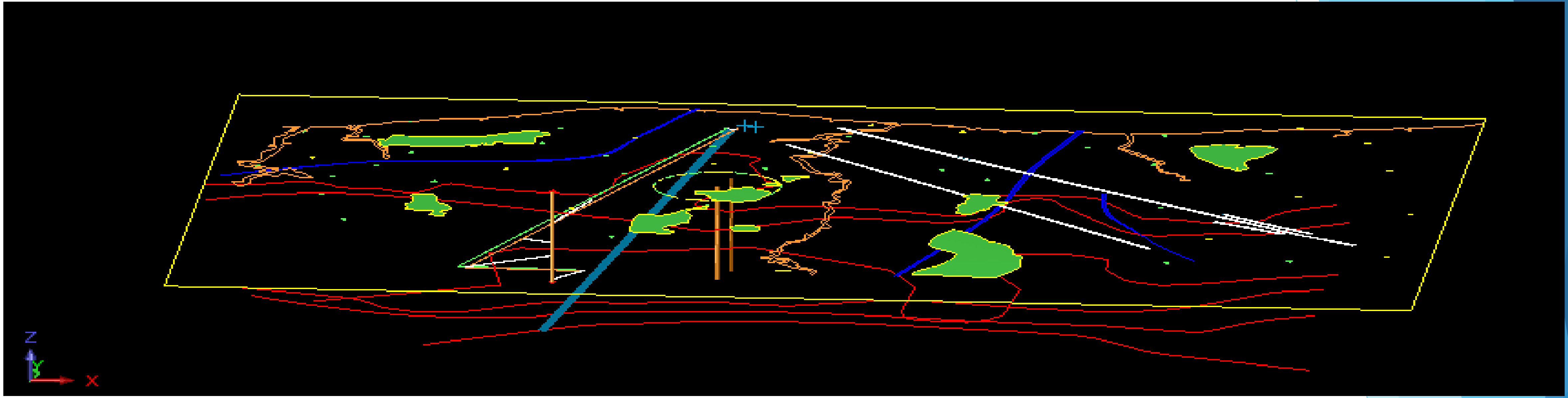
# Mining and its Associations

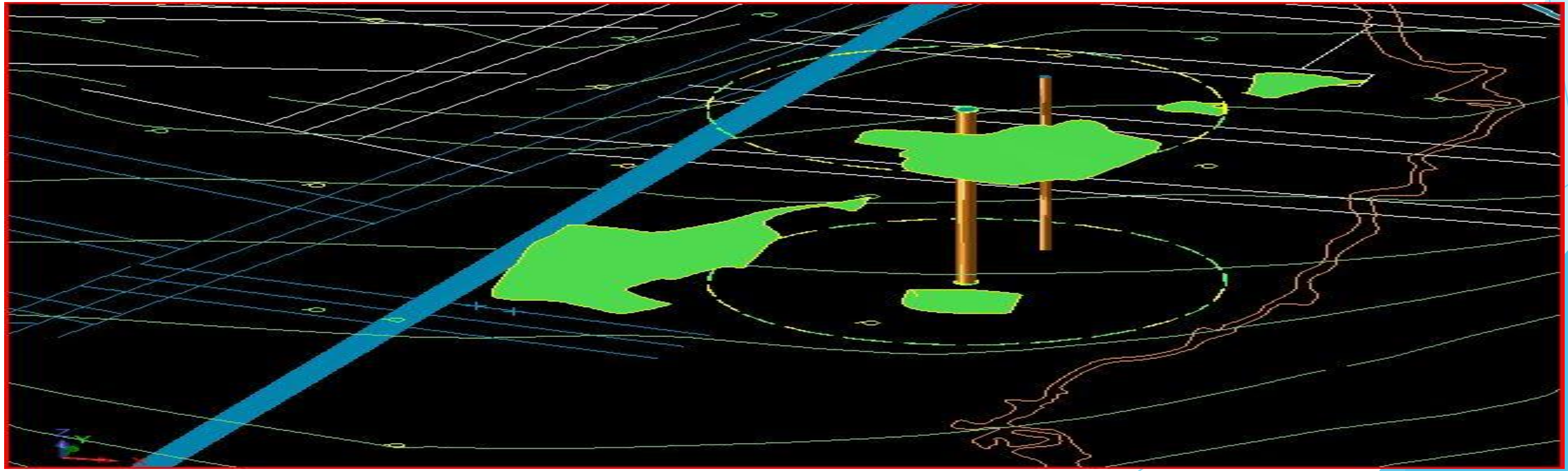


# Mine planning and its importance

- Planning is basically the process of taking decisions, on the basis of available information, to fulfill an objective while satisfying the constraints present. It involves in identifying (or developing) the possible alternatives and selecting the optimum one through proper evaluation, and the process is generally a reiterative one. Design is the process of translating the planning decisions into tangible form.
- In mine planning stage, the decisions, mostly financial, are taken to optimize an objective function (net present value, total profit, internal rate of return, total conservation, etc.) depending on the requirement on the basis of available exploration data and other information while satisfying the various constraints (e.g, available capital, legal considerations, market conditions, physical and biological environment, techno-economic environment, climatic conditions, geographic location, site characteristics, deposit characteristics, etc.) which are likely to affect the venture.
- In mine design stage, the planning decisions are translated in the form of maps, drawings, figures, charts, tables, etc.

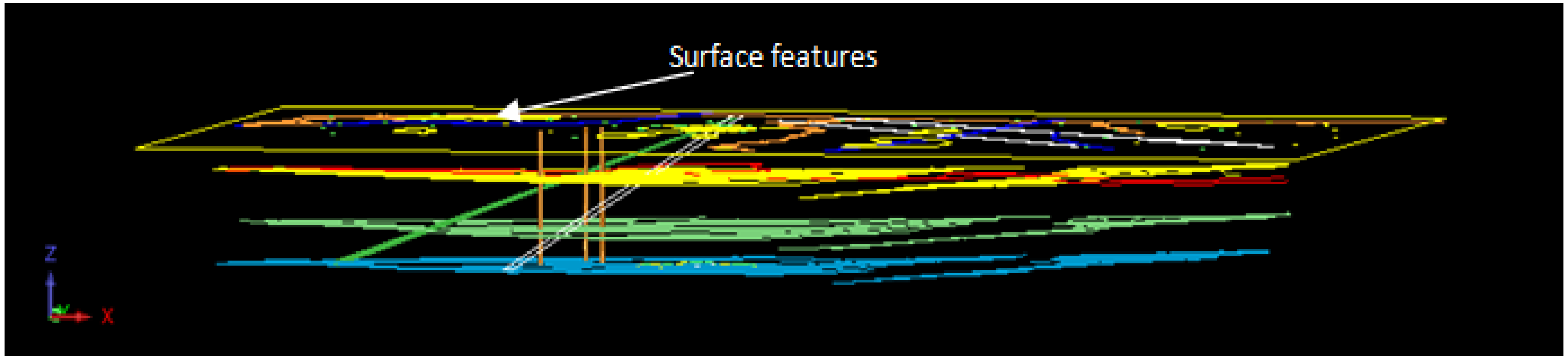
# Mine design exercises

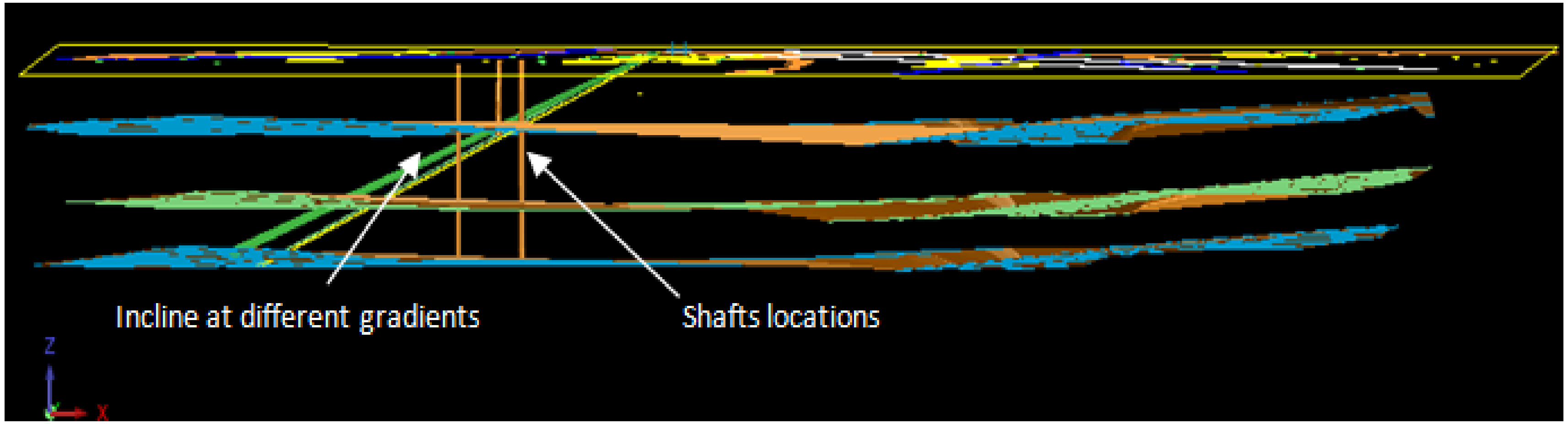


Mine planning helps to reduce the following risks

- Mineral risk
- Promoters and sponsors risk
- Project completion risk
- Capital cost and delay risk
- Operation risk
- Market risk
- Sovereign risk such as economic environment, political and regulatory environment and foreign currency risk



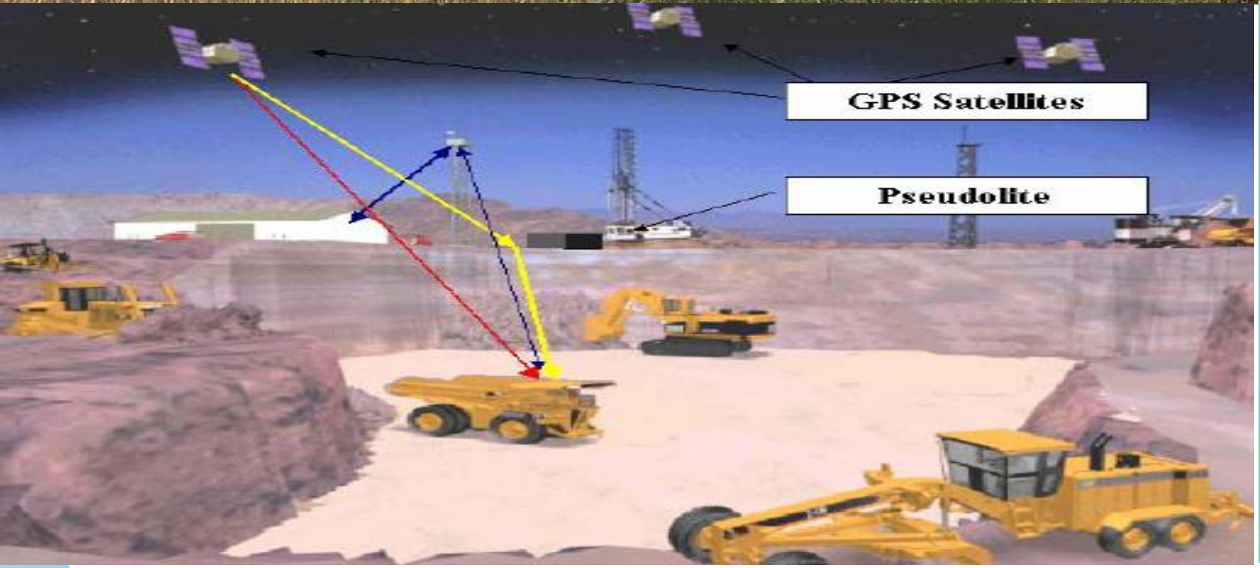

Study of various options for mine design

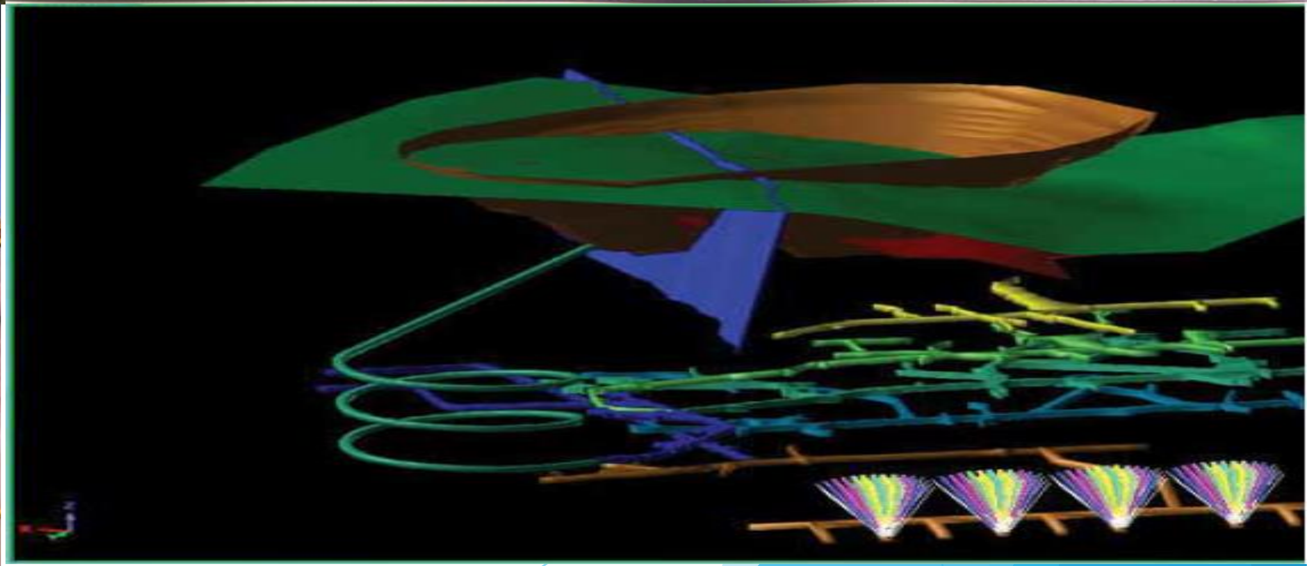
# Components of mine planning

- *Selection of break-even cut-off grade*
- *Development of the 3-D economic block model*
- *Choice of mining method*
- *Identification of geometrical extents of mine*
- *Estimation of total and mineable ore reserve vis-à-vis grade*
- *Selection of optimum mine and mill cut-off grades in case of surface mines*
- *Estimation of targeted production and mine life*
- *Planning ore and waste excavation schedule vis-à-vis intermediate pit layout in case of surface mines*
- *Planning mine development and ore excavation schedule in case of underground mines*
- *Planning the equipment system*
- *Calendar planning*

- *Slope stability / excavation stability planning*
- *Waste dump planning (in case of surface mines)*
- *Tailings dump planning (in case of metal mines)*
- *Washery rejects dump planning (in case of coal mines)*
- *Mine drainage planning*
- *Mine transportation system planning*
- *Mine illumination system planning*
- *Mine ventilation system planning (in case of underground mines)*
- *Environmental management planning*
- *Environmental disaster management planning*
- *Mine reclamation/mine closure planning*


11/01/2012

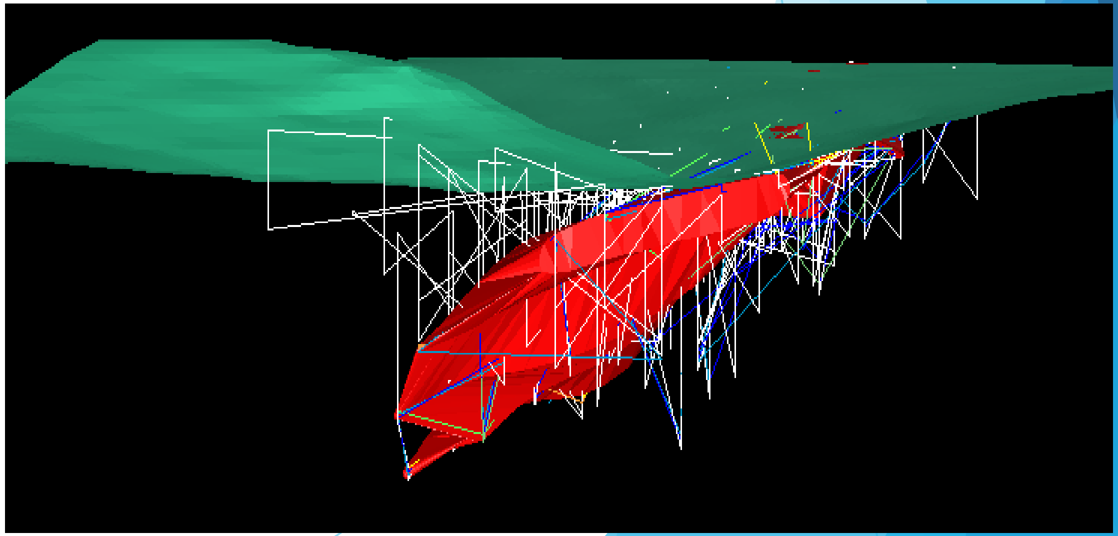

GPS Satellites
Pseudolite



# Technical and economic information for planning

- Topography and contour
- Climatic conditions
- Surface
- Water (potable and process)
- Mine Water as determined by prospect holes
- Ore body characteristics
- Host rock characteristics
- Ground water data
- Locations for concentrator/ coal washery – factors to consider for optimum location
- Tailing pond area
- Roads

- Power
- Smelting (in case of metal mines)
- Land ownership
- Government



- **Economic climate**

Principal industries, availability of labor and normal work schedules, wage scales, tax structure, availability of goods and services (housing, stores, recreation, medical facilities and unusual local disease, hospital, schools, etc.), material costs and/or availability (fuel oil, concrete, gravel, borrow material for dams, etc.), purchasing.

- **Waste dump location** (in case of surface mines)

Physico-mechanical properties of dump foundation, haul distance, haul profile, amenable to future leaching operation.

- **Mill Tailings dump location** (in case of metal mines)
- Physico-mechanical properties of dump foundation, haul distance, haul profile,
- **Washery Rejects dump location** (in case of coal mines)
- Physico-mechanical properties of dump foundation, haul distance, haul profile.
- **Accessibility of principal town to outside**

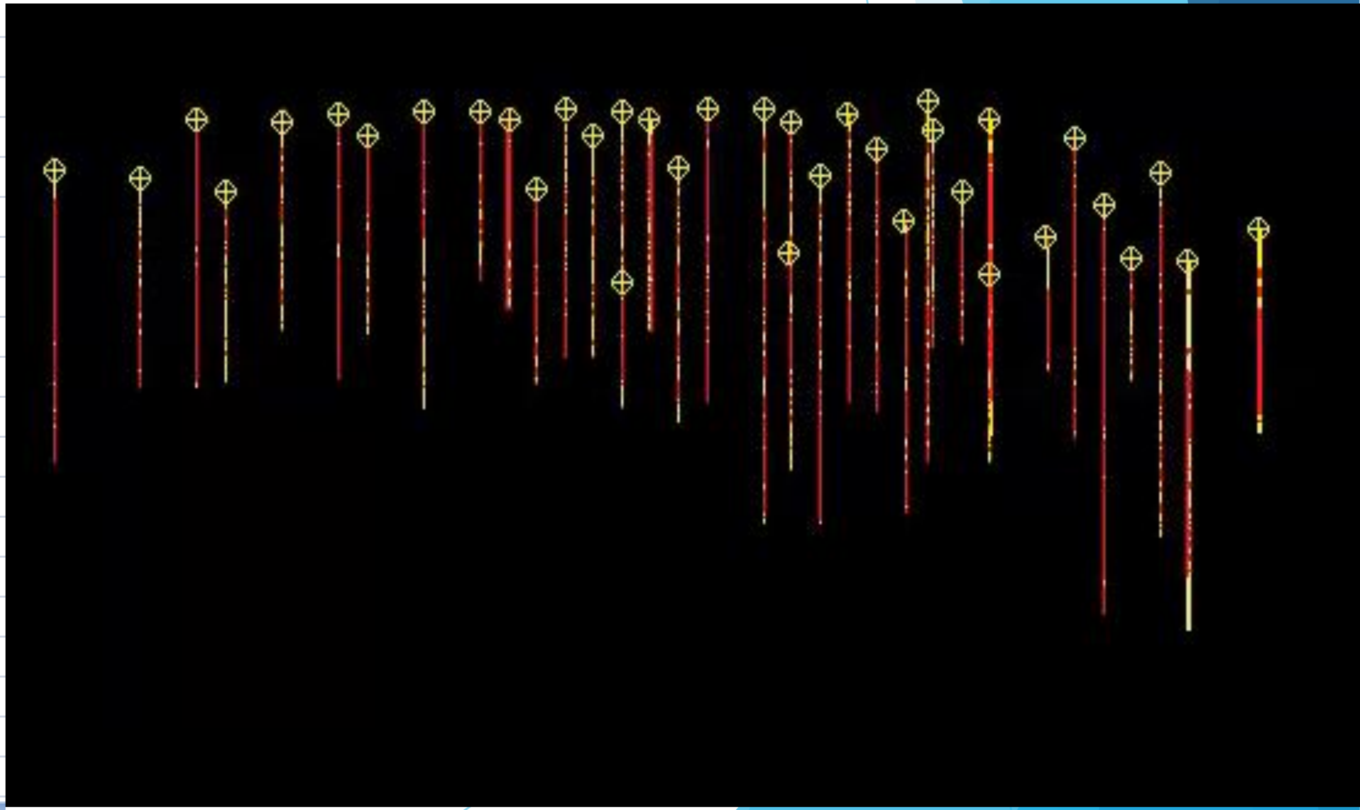
Methods of transportation available, reliability of transportation available, communications, etc.

# Exploratory drills: Exploratory drilling techniques.pdf

- Auger drilling
- Percussion rotary air blast drilling (RAB)
- Air core drilling
- Cable tool drilling
- Reverse circulation (RC) drilling
- Odex drilling
- Diamond core drilling

# Interpretation of bore hole data

| hole_id | Sample_ic | depth_fro | depth_to | fe | sio2 | al2o3 | loi | p |
|---|---|---|---|---|---|---|---|---|
| B01 | | 0 | 2.44 | 61.6 | 2.5 | 5.01 | 3.9 | 0.06 |
| B01 | | 2.44 | 4.88 | 59.8 | 4.4 | 5.39 | 4.2 | 0.06 |
| B01 | | 4.88 | 6.4 | 61 | 4 | 5.27 | 3 | 0.06 |
| B01 | | 6.4 | 9.75 | 56.4 | 5.2 | 7.45 | 5.2 | 0.06 |
| B01 | | 9.75 | 11.89 | 63.8 | 2.2 | 2.4 | 3.8 | 0.04 |
| B01 | | 11.89 | 13.42 | 63.8 | 1.6 | 3.37 | 3.3 | 0.06 |
| B01 | | 13.42 | 16.17 | 63.9 | 1.8 | 3.52 | 2.8 | 0.06 |
| B01 | | 16.17 | 18.91 | 64.1 | 1.2 | 3.9 | 2.7 | 0.06 |
| B01 | | 18.91 | 20.43 | 64.8 | 2.6 | 2.8 | 2.5 | 0.06 |
| B01 | | 20.43 | 22.26 | 63 | 2 | 4.5 | 2.9 | 0.06 |
| B01 | | 22.26 | 24.4 | 66.2 | 2 | 2 | 2 | 0.04 |
| B01 | | 24.4 | 26.23 | 65 | 2.68 | 2.02 | 2.1 | 0.06 |
| B01 | | 26.23 | 28.67 | 66 | 1.58 | 2.14 | 1.8 | 0.04 |
| B01 | | 28.67 | 30.5 | 66.2 | 1.6 | 2.04 | 1.6 | 0.04 |
| B01 | | 30.5 | 32.33 | 62.8 | 4.3 | 2.9 | 2.5 | 0.06 |
| B01 | | 32.33 | 34.16 | 64.6 | 3.16 | 2.36 | 2 | 0.06 |
| B01 | | 34.16 | 36.59 | 64.8 | 4.46 | 1.11 | 1.65 | 0.05 |
| B01 | | 36.59 | 39.02 | 64.2 | 5.16 | 1.22 | 1.71 | 0.06 |
| B01 | | 39.02 | 41.44 | 65.6 | 2 | 2.07 | 1.9 | 0.05 |
| B01 | | 41.44 | 43.24 | 66 | 2.6 | 1 | 1.9 | 0.04 |
| B01 | | 43.24 | 45.06 | 64.4 | 2.84 | 1.23 | 3.6 | 0.04 |
| B01 | | 45.06 | 46.86 | 64.1 | 3.5 | 1.29 | 3.3 | 0.04 |
| B01 | | 46.86 | 48.96 | 64.4 | 6.12 | 2.21 | 3.2 | 0.04 |
| B01 | | 48.96 | 51.06 | 64 | 4.16 | 1.2 | 2.85 | 0.06 |



# Thank you